its father. The Rh negative mother becomes sensitized by carrying Rh positive foetus within her body. Due to damage of blood vessels, during child birth, the mother's immune system recognizes the Rh antigens and gets sensitized. The sensitized mother produces Rh antibodies. The antibodies are IgG type which are small and can cross placenta and enter the foetal circulation. By the time the mother gets sensitized and produce anti 'D' antibodies, the child is delivered.

Usually no effects are associated with exposure of the mother to Rh positive antigen during the first child birth, subsequent Rh positive children carried by the same mother, may be exposed to antibodies produced by the mother against Rh antigen, which are carried across the placenta into the foetal blood circulation. This causes haemolysis of foetal RBCs resulting in haemolytic jaundice and anaemia. This condition is known as **Erythoblastosis foetalis or Haemolytic disease of the new born (HDN).**

Prevention of Erythroblastosis foetalis

If the mother is Rh negative and foetus is Rh positive, anti D antibodies should be administered to the mother at 28th and 34th week of gestation as a prophylactic measure. If the Rh negative mother delivers Rh positive child then anti D antibodies should be administered to the mother soon after delivery. This develops passive immunity and prevents the formation of anti D antibodies in the mothers blood by destroying the Rh foetal RBC before the mother's immune system is sensitized. This has to be done whenever the woman attains pregnancy.

4.4 Sex Determination

Sex determination is the method by which the distinction between male and female is established in a species. Sex chromosomes determine the sex of the individual in dioecious or unisexual organisms. The chromosomes other than the sex chromosomes of an individual are called autosomes. Sex chromosomes may be similar (homomorphic) in one sex and dissimilar (heteromorphic) in the other. Individuals having homomorphic sex chromosomes produce only one type of gametes (homogametic) whereas heteromorphic individuals produce two types of gametes (heterogametic).



Y CHROMOSOME

The human Y chromosome is only 60 Mb in size with 60 functional genes whereas X chromosomes are 165 Mb in size with about 1,000 genes.

Chromosomal basis of sex determination



Heterogametic Sex Determination:

In heterogametic sex determination one of the sexes produces similar gametes and the other sex produces dissimilar gametes. The sex of the offspring is determined at the time of fertilization.

Heterogametic Males

In this method of sex determination the males are heterogametic producing dissimilar gametes while females are homogametic producing similar gametes. It is of two kinds XX-XO type (e.g. Bugs, cockroaches and grasshoppers) and XX-XY type (e.g. Human beings and *Drosophila)*.

Heterogametic Females

In this method of sex determination the females are heterogametic producing dissimilar gametes while males are homogametic producing similar gametes. To avoid confusion with the XX-XO and XX-XY types

TN_GOVT_XII_Zoology_chapter 4.indd 48 24-02-2020 4.01.10 PM